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WEEK :-1

DAY:-6

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Pre-Dataset Readiness + Mentor Review

1. Literature Readiness

- The literature review was completed to understand existing approaches for battery State of Charge estimation and Remaining Charging Time prediction. · The reviewed studies show that Machine Learning methods such as Linear Regression, Decision Trees, Random Forest, Support Vector Regression and Gradient Boosting can be used for battery-related prediction tasks.
- The literature also indicates that battery voltage, current, temperature, charging power and time are important variables for battery prediction.
- Based on the literature review, both simple baseline models and more advanced Machine Learning models will be considered.

2. Data Readiness

Before modeling, the supplied dataset will be checked for:

- Number of rows and columns

- Available variables
- Data types
- Missing values
- Invalid values
- Duplicate observations
- SOC range
- Remaining Charging Time range
- Timestamp structure
- Charging sessions/cycles
- Sampling interval
- Possible data leakage
- The original raw dataset will be preserved without modification.

3. Proposed Preprocessing Plan

The proposed preprocessing steps are:

1. Load and inspect the supplied dataset.
2. Preserve the original raw dataset.

3. Create a separate working copy.
4. Check data types and column names.
5. Handle timestamps appropriately.
6. Check missing and invalid values.
7. Check duplicate observations.
8. Identify charging sessions or battery cycles.
9. Encode categorical variables such as charging status if required.
10. Apply feature scaling where required by the selected model.
11. Perform feature engineering only when it does not introduce future information.
12. Document every preprocessing step.

4. Proposed Input Features

The expected input features are:

- Time / Timestamp
- Voltage
- Current
- Power

- Temperature
- Charging Status

Other relevant variables available in the supplied dataset

The final feature list will be confirmed after dataset inspection.

5. Target Variables

- Target 1 — State of Charge (SOC)

SOC will be predicted as a continuous numerical value.

- Target 2 — Remaining Charging Time (RCT)

RCT will be predicted as a continuous numerical value.

If RCT is not directly provided in the dataset, it may be calculated from timestamps after confirming the definition of the charging endpoint.

6. Proposed Modeling Plan

The following models are proposed:

- Baseline Models
 - Linear Regression

- Machine Learning Models
 - Decision Tree Regression
 - Random Forest Regression
 - Support Vector Regression
 - Gradient Boosting / XGBoost

- Optional Model
 - ANN may be considered if the dataset is sufficiently large and contains meaningful sequential information.

7. Proposed Evaluation Plan

SOC and RCT will be evaluated separately.

The proposed metrics are:

- MAE — Mean Absolute Error
 - Measures the average absolute difference between actual and predicted values .Lower MAE indicates better performance.

- RMSE — Root Mean Squared Error
 - Gives greater importance to larger prediction errors.Lower RMSE indicates better performance.

- R^2 — Coefficient of Determination

- Measures how well the model explains variation in the target. Higher R^2 generally indicates better performance.

8. Proposed Data Split Strategy

The proposed split is approximately:

- Training: 70%
- Validation: 15%
- Testing: 15%

The exact proportions may be adjusted according to the dataset size. The same charging session should not appear in both training and testing data.

If a session/cycle identifier is not available, a suitable time-aware splitting method will be considered.

9. Leakage Prevention

The following rules will be followed:

- Future measurements will not be used as input features.
- Target values will not be used as input features.

- Information from the end of a charging session will not be used when it would not be available at prediction time.
- Preprocessing parameters will be learned only from training data. · The same charging session should not be distributed across training and testing sets.
- Features based on future observations will not be created.

10. Experiment Plan

- 01 | SOC | Linear Regression | Simple baseline |
- 02 | SOC | Decision Tree | Nonlinear model |
- 03 | SOC | Random Forest | Ensemble model |
- 04 | SOC | Gradient Boosting/XGBoost | Advanced model |
- 05 RCT | Linear Regression | Simple baseline |
- 06 | RCT | Decision Tree | Nonlinear model |
- 07 | RCT | Random Forest | Ensemble model |
- 08 | RCT | Gradient Boosting/XGBoost | Advanced model |

11. Data-Readiness Checklist

- Raw dataset preserved

- Missing values checked
- Invalid values checked
- Duplicate rows checked
- SOC range checked
- RCT range checked
- Timestamp structure checked
- Leakage risks checked
- Train/validation/test strategy documented

12. Mentor Discussion Points

The following points will be confirmed with the mentor:

1. Final definition of Remaining Charging Time.
2. Definition of the charging endpoint.
3. Final input features.
4. Data splitting strategy.

5. Evaluation metrics.

6. Handling of missing and invalid values.

7. Whether session/cycle-based splitting is possible. 8. Whether

ANN/LSTM is necessary or appropriate. 9. Any project-specific

constraints on preprocessing or modeling.